

SIMATS ENGINEERING ASSESSMENT TOOL 1

COURSE NAME: Cryptography and Network Security
COURSE CODE: CSA5102

COURSE OUTCOME COVERED

CO1: Apply classical encryption techniques using symmetric and asymmetric ciphers to encrypt and decrypt data for the ensuring data security. (BL3, BL4)

Assessment Tool Weightage: 40%

QUESTIONS: 5

Total Marks:50

Annexure A

APPLICATION-BASED QUESTIONS

Code of Conduct:

I v.sree Ruthin Reddy (192324112) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

v. Ruthin

Signature of the student:

98
100
P

Annexure A

APPLICATION-BASED QUESTIONS

1. Scenario: Event Communication Security: A student organization is preparing to share confidential event details with its core members using a simple classical encryption method. During testing, a message "MEET AT NOON" is transformed into an encoded version using a fixed shift pattern.

- a. Reconstruct how the encrypted message might appear if a shift of 4 positions is applied.
- b. Explain how an authorized member, aware of the encoding rule, would retrieve the original message.

2. Scenario: Intercepted Suspicious Message: A cybersecurity intern intercepts the coded message: "GSRH RH Z HVXIVG" during network monitoring. The structure of the text suggests a simple substitution mechanism rather than complex encryption.

- a. Deduce the most likely original message.
- b. Justify your answer by analysing repeating patterns, letter structures, or known cipher characteristics.

3. Scenario: Secure Messaging System: A company deploys a keyword-based encryption technique for internal communication. An employee sends the word "HELLO" using the keyword "KEY" to ensure confidentiality.

- a. Demonstrate how the encryption process transforms the message step-by-step.
- b. Explain how the repeating keyword influences each character of the ciphertext.

4. Scenario: Suspicious Digital Asset Investigation: During a digital forensic audit, investigators suspect that a company logo file may contain hidden confidential information embedded within it. The file appears normal upon casual inspection.

- a. Analyze how investigators could detect the presence of hidden data within the image.
- b. Recommend two techniques that could strengthen the security of such hidden communications against detection or extraction.

5. Scenario: Legacy Encryption in Corporate Communication: A legacy system in a company still uses a transposition-based encryption method with three levels of arrangement. A received message appears as: "WECRLTEERDSOEFEAOCAIVDEN".

- a. Determine the most probable original message by reversing the encryption process.
- b. Critically evaluate why such a method is vulnerable in modern communication systems.
- c. Propose one practical enhancement to improve the confidentiality of messages in this scenario.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Concepts	25 23	Demonstrates thorough understanding and effectively integrates multiple concepts/technologies	Good understanding with proper integration of most concepts	Basic understanding; limited or partial integration	Poor understanding; unable to integrate concepts
Experimental Design	30 30	Well-planned, systematic implementation with optimal solution	Correct implementation with minor issues	Basic implementation with errors or inefficiencies	Incorrect or incomplete implementation
Use of Tools	20 20	Effective and advanced use of tools/technologies with proper configuration	Appropriate use of tools with correct execution	Limited or inconsistent use of tools	Incorrect or minimal use of tools
Analysis of Results	15 15	Insightful analysis with accurate interpretation and validation of results	Good analysis with reasonable interpretation	Basic analysis with limited insights	Poor or incorrect analysis of results
Documentation	10 10	Clear, well-structured documentation with diagrams, code, and results	Organized documentation with necessary details	Basic documentation with missing details	Poor or incomplete documentation

$$\frac{98}{100}$$

1 (a) Scenario: Secure messaging system

A company uses a keyword-based encryption technique to protect internal communication. An employee wants to send the message "Hello" using the keyword "key".

This is an example of Vignere cipher, where the letters of the keyword are repeatedly used to encrypt the letters of the plaintext.

(a) demonstrate the encryption process step-by-step.

We use the standard alphabet numbering

A = 0, B = 1, C = 2, ..., Z = 25

The encryption formula is:

$$C = (P + K) \bmod 26.$$

where:

- P = numerical value of plaintext letter.
- K = numerical value of keyword letter
- C = numerical value of ciphertext letter.

Step ①: write the plaintext

HELLO

Step ②: Repeat the keyword to match the message length.

keyword: key

Since the message has 5 letters, repeat the keyword.

Position	1	2	3	4	5
Plain text	H	E	L	L	O
Key word	K	E	Y	K	E

step (3): Convert letters into numbers.

Letter	H	E	L	L	O
Plain text	7	4	11	11	14
Key word	K	E	Y	K	E
Key word value.	10	4	24	10	4

step (4): perform the encryption.

using:

$$C = (P + K) \bmod 26$$

1. $H + K$

$$(7 + 10) \bmod 26 = 17$$

$$17 = R$$

2. $E + E$

$$(4 + 4) \bmod 26 = 8$$

$$8 = I$$

3. $L + Y$

$$(11 + 24) \bmod 26 = 9$$

$$9 = J$$

4. $L + K$

$$(11 + 10) \bmod 26 = 21$$

$$21 = V$$

5. $O + E$

$$(11 + 4) \bmod 26 = 18$$

$$18 = S$$

Therefore,

HELLO $\rightarrow$ RIJVS

Complete encryption table.

Position	P. T	P. value	Keyword	K. value	Cipher Text.
1	H	7	K	10	R
2	E	4	E	4	I
3	L	11	Y	24	J
4	L	11	K	10	V
5	O	14	E	4	S

Final answer:

Plain text: Hello

Keyword: key

Ciphertext: R I J V S.

1 (b) How does the repeating keyword influence each ciphertext character?

The keyword determines how much each plaintext letter is shifted.

Here, the keyword is only three letters long:

K $\rightarrow$ shift by 10

E $\rightarrow$ shift by 4

Y $\rightarrow$ shift by 24

After using K, E, Y the keyword starts again:

K $\rightarrow$ E

10 $\rightarrow$ 4 $\rightarrow$ 24 $\rightarrow$ 10 $\rightarrow$ 4

For example,

• First $L + Y \rightarrow J$

• Second $L + K \rightarrow V$

So, the repeating keyword makes the encryption stronger than a simple Caesar cipher because different positions use different shifts.

2(a) SCENARIO: SUSPICIOUS DIGITAL ASSET INVESTIGATION.

A company logo appears completely normal, but investigators suspect that confidential information has been hidden inside the image.

This technique is commonly called steganography.

What is steganography?

Steganography is a technique of hiding secret information inside an apparently ordinary file such as:

- Image
- Audio
- Video
- Document

The main objective is to make the existence of the secret message difficult to notice.

How can investigators detect hidden data?

Investigators should use a combination of visual inspection, metadata analysis, statistical analysis, and specialised forensic tools.

Examine the image visually:

- Strange color patterns.
- Unusual pixels
- Distorted noise
- Color variations.

check the metadata:

- File creation date
- modification data
- software used to create the file
- Image dimensions
- camera information
- Author information
- comment
- embedded descripting.

compare with the original image:

They can compare:

- File size
- dimension
- color information
- pixel values
- compression characteristics.
- metadata
- Hash values.

Analyse the image's pixel data:

one common image-steganography approach hides information in the least significant bits.

a pixel may contain binary values such as:

10110110

changing the last bit:

10110110 $\rightarrow$ 10110111

They may look for:

- non random patterns.
- repeated binary sequences.
- embedded text
- unexpected statistical distributions.

(b) Two technical to strengthen hidden communications against detection on extraction.

Two techniques can make the hidden communication more resistant to analysis.

Technique ①: encrypt the secret message before embedding.

original message:

CONFIDENTIAL DATA

↓

encryption

8FJA....X72K

↓
embedded encrypted data into image

layer ①: steganography

Hides the existence of the message

layer ②: encryption

Hides the actual content if the hidden data is extracted.

Technique ②:

the goal is reduce:

- visible artifacts.
- abnormal pixel patterns.
- statistical anomaly.
- detectable changes in color distribution.